

Congenital Cytomegalovirus Infection - Lessons from a Clinical Case.

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Cytomegalovirus (CMV) is a worldwide spread herpes virus that establishes a latent infection after the primary infection. It becomes a major problem in immunocompromised patients and in cases of primary or reactivated infection during pregnancy. CMV is the most common congenital infection and is the leading infectious cause of sensorineural deafness and cerebral mental retardation. To raise the attention to the discordance in our knowledge of cCMV infection and the implementation of the knowledge in practice. CMV-DNA was extracted from saliva and plasma samples according to the protocol of DNA-Sorb-A, Sacace, Biotechnologies, Italy. Polymerase chain reaction (PCR) was performed using a commercial kit based on the Taq-man principle: Quantitative RT-PCR for CMV-DNA detection (Sacace, Biotechnologies, Italy). The linear range of the CMV Real-TM Quant PCR test is 500-10000000 copies/ml, and the reported sensitivity is 400 copies/ml. The infant was born with clinical manifestations involving development of hepatosplenomegaly, chorioretinitis, anemia, microcephaly and simultaneous dilatation of the brain ventricles. CMV infection was confirmed using modern PCR studies. This case highlights the need for specific virological/PCR tests to be performed for all children with the least suspected congenital infection, especially when there is an option of a specific treatment.

Effectiveness and cost benefit of a proposed live cytomegalovirus vaccine in the prevention of congenital disease.

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Congenital infection by human cytomegalovirus (CMV) is presently the leading infectious cause of mental retardation and congenital deafness in the United States. Live CMV vaccines in healthy adults have been shown to be safe and to induce immune responses similar to those that occur with natural CMV infection. Yet, only recently has a live CMV vaccine been tested for its protective ability. To evaluate the cost benefit and effectiveness of the proposed live CMV vaccine, we compared the following strategies: routine immunization, selective immunization of those women screened and found to be seronegative, and no immunization. Our results show that, when direct costs alone are considered, routine immunization of healthy women aged 15-25 years is cost beneficial even in populations with CMV seroprevalence as high as 87%. In populations with lower seroprevalence (55%-70%), for every 100,000 women immunized, more than 24 cases of symptomatic congenital CMV infection at birth and a similar number of cases with late sequelae (mainly deafness) would be prevented yearly. Such immunization would result in a net annual saving of \$2.5 million.

Mother-to-fetus transmission of cytomegalovirus. A review.

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Congenital human cytomegalovirus (CMV) infection is the leading infectious cause of mental retardation, sensorineural deafness and visual impairment. It is mainly related to a primary maternal infection. The placenta should be considered the most important site of both the protection of the fetus from CMV infection and the transmission of CMV from mother to fetus. The control of the passage of CMV across the placenta probably involves a cascade of regulatory events. Roles are played by factors relating to the

host immune-selective pressures, such as local cytokines and maternal CMV-specific neutralizing antibodies. The presence of other pathogens at the maternal-fetal interface also influences the outcome of CMV infection. Further investigations are needed in which clinical CMV strains are applied in in vitro studies to unravel the molecular mechanism of the intrauterine transmission of CMV and to elucidate the complex regulation that leads to prevention of the in utero transmission of CMV in vivo.

Role of cytomegalovirus in sensorineural hearing loss of children: a case-control study Tehran, Iran.

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Congenital CMV is the most common cause of nonhereditary sensorineural hearing loss in children. More than 40% of deafness cases with an unknown cause are due to congenital CMV. When CMV infection is diagnosed in the first year of life, treatment with gancyclovir is recommended and is associated with diminished occurrence of SNHL. Previous studies in Tehran showed congenital CMV infection developed in 2.6% of neonates. To determine the role of cytomegalovirus in sensorineural hearing loss in children by comparing the CMV antibodies quantitatively between SNHL and controls. This case-control study was done in the ENT Department of Hazrat Rasoul Akram Hospital in Tehran (2002-2003). This study was carried out based on diagnostic parameters (AAO criteria) for SNHL and a healthy control. We compared the specific cytomegalovirus antibodies (IgM and IgG measured by ELISA method) in 95 blood samples of infants with SNHL (mean age=35+/-30 months) and 63 healthy matched infants (mean age=38.7+/-27.3 months) as controls. 59.6% of cases were male; 40.4% were female. Acute and previous immunity to cytomegalovirus (IgM and IgG) was found in 34.7% and 72.6% of SNHL children, acute infection detected in 3.5% of controls, previous immunity (IgG) detected in 94.7% of them. Acute CMV infection was higher in the cases (P-value=0.000) but previous immunity (CMV-IgG) was higher in the controls (P-value=0.001). Mean age of cases with acute infection (CMV-IgM) was 40 months and for previous immunity (CMV-IgG) was 35 months in SNHL cases and 40 months in controls, respectively. Cytomegalovirus is one of the most common infectious agents in SNHL children compared to the healthy children. Probably both congenital and acquired CMV can induce progressive hearing loss in our cases. We prefer at least in our country to consider seropositive (CMV-IgM) SNHL children (less than 1 year old) as congenital form. But we are not able to differentiate the congenital from the acquired infection in seropositive (CMV-IgM) SNHL children after first year of life. It should be subjected to randomized controlled trial for treatment of acquired type of CMV infection in SNHL children with ganciclovir.

Decoding human cytomegalovirus for the development of innovative diagnostics to detect congenital infection.

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Cytomegalovirus is the most common cause of congenital infectious disease and the leading nongenetic etiology of sensorineural hearing loss. Although most infected neonates are asymptomatic at birth, congenital cytomegalovirus infection is responsible for nearly 400 infant deaths annually in the United States and may lead to significant long-term neurodevelopmental impairments in survivors. The resulting financial and social burdens of congenital cytomegalovirus infection have led many medical centers to initiate targeted testing after birth, with a growing advocacy to advance universal newborn screening.

While no cures or vaccines are currently available to eliminate or prevent cytomegalovirus infection, much has been learned over the last five years regarding disease pathophysiology and viral replication cycles that may enable the development of innovative diagnostics and therapeutics. This Review will detail our current understanding of congenital cytomegalovirus infection, while focusing our discussion on routine and emerging diagnostics for viral detection, quantification, and long-term prognostication. **IMPACT:** This review highlights our current understanding of the fetal transmission of human cytomegalovirus. It details clinical signs and physical findings of congenital cytomegalovirus infection. This submission discusses currently available cytomegalovirus diagnostics and introduces emerging platforms that promise improved sensitivity, specificity, limit of detection, viral quantification, detection of genomic antiviral resistance, and infection staging (primary, latency, reactivation, reinfection).

Long-term follow-up of a series of 24 congenital CMV-infected babies with false negative amniocentesis.

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Congenital CMV infection is the most common congenital infection worldwide and a major cause of neurological impairment and sensorineural hearing loss. Fetal CMV infection is confirmed by a positive PCR test in the amniotic fluid (amniocentesis performed after 18-20 weeks of gestation and at least 8 weeks after maternal infection). However, despite a negative antenatal CMV PCR result, some newborns can be tested positive at birth. Although not widely documented, the prognosis for these babies appears to be good. The aim of this study is to evaluate the long-term prognosis of fetuses with a false-negative AFS for cCMV, with a minimum follow-up period of 6 years. This is a retrospective cohort study of false-negative amniocentesis reported at the CUB-Hôpital Erasme and Hôpital CHIREC in Brussels between 1985 and 2017. Of the 712 negative CMV PCR amniocenteses, 24 had a CMV PCR positive at birth. The false negative rate was 8.6 %. Of the 24 cases, 9 primary maternal infections occurred in the first trimester, 14 in the second trimester and 1 in the third trimester. Among the 24 children, 2 had symptoms at birth (hyperbilirubinemia and left paraventricular cysts), but all had normal follow-up (minimum 4 years, mean 16,6 years). Only 2 cases could be explained by early amniocentesis. Among the others, the false-negative results could be attributed to a low viral load, a delayed infection or, less likely, to a sample degradation. Despite the false-negative results, all 24 children had a normal long-term follow-up.

Congenital Cytomegalovirus Severity Definitions and Treatment Decisions around the World: A Systematic Scoping Review of the Literature.

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Congenital cytomegalovirus (cCMV) is the most common cause of congenital infection and the leading cause of non-genetic sensorineural hearing loss in childhood. While treatment trials have been conducted in symptomatic children, defining asymptomatic infection can be complex. We performed a scoping review to understand how infection severity is defined and treated globally, as well as the various indications for initiating treatment. We conducted an electronic search of MEDLINE, EMBASE, Scopus, and the Cochrane Library, using combinations of the following terms: "newborn", "baby", "child", "ganciclovir", "valganciclovir", and "cytomegalovirus" or "CMV". We included eligible prospective and

retrospective studies, case series, and randomized clinical trials (RCTs) published up to May 2024. A total of 26 studies were included, of which only 5 were RCTs. There was significant heterogeneity between studies. The most commonly considered criteria for symptomatic infection were microcephaly (23/24 studies), abnormal neuroimaging (22/24 studies), chorioretinitis/ocular impairment (21/24 studies), and hearing impairment (20/24 studies). Two studies also included asymptomatic newborns in their treatment protocols. Outcome measures varied widely, focusing either on different hearing assessments or neurocognitive issues. Our literature analysis revealed significant variability and heterogeneity in the definition of symptomatic cCMV infection and, consequently, in treatment approaches. A consensus on core outcomes and well-conducted RCTs are needed to establish treatment protocols for specific groups of newborns with varying manifestations of cCMV.

Treatment of congenital cytomegalovirus infection.

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Congenital cytomegalovirus (CMV) is the most common cause of congenital infection worldwide, the most common nongenetic cause of sensorineural hearing loss in children, and a cause of neurodevelopmental disorders in the brain. Infants with symptomatic congenital CMV infection may benefit from hearing and neurodevelopmental outcomes, particularly if antiviral treatment is initiated within the first month of life. Infants with life-threatening symptoms are recommended to receive 2-6 weeks of intravenous ganciclovir and then switch to oral valganciclovir, and those without life-threatening symptoms are recommended to use oral valganciclovir during the entire 6-month period. During antiviral drug treatment, absolute neutrophil count, platelet count, blood urea nitrogen, creatinine, and liver function tests were performed to identify neutropenia, thrombocytopenia, renal failure, and liver failure. This review investigated the evidence to date of treating congenital CMV infection.

Cytomegalovirus seronegativity rate in pregnant women and primary cytomegalovirus infection during pregnancy in rural Germany.

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Congenital cytomegalovirus (CMV) infection is the most common congenital infection worldwide and one of the leading causes of congenital hearing loss in newborns. The aim of this study was to determine the seroprevalence rate for cytomegalovirus in pregnant women and the rate of CMV serological testing utilised during pregnancy in a rural region in Germany. Retrospective data on the prevalence of CMV IgG and IgM antibodies were obtained from 3,800 women, identified in the study group of 19,511 pregnant women from outpatient settings whose samples were collected between 1 and 2014 and 30 April 2018. In addition, the serological CMV status in regards to various billing methods was further analyzed. Serological CMV tests were performed in 3,800 (19.5%) out of 19,511 pregnant women. 2,081 (54.8%) of these women were CMV seronegative. Among those, seroconversion rate of 0.37-1.42% was identified. A proportion of 2,710 (14.7%) of all 18,460 women with statutory health insurance made use of the CMV testing as an individual health service. The low uptake of CMV serological testing in the study population covered indicates low risk awareness among pregnant women and their healthcare professionals. Presented seronegativity rates and routine seroconversion rate, demonstrate importance to improve intervention strategy to prevent fetomaternal CMV transmission.

Performance of Targeted Congenital Cytomegalovirus Screening in Newborns Failing Universal Hearing Screening: A Multicenter Study.

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Cytomegalovirus (CMV) is the most frequent cause of congenital infection and $\approx 20\%$ of all infected neonates present or will develop sensorineural hearing loss. Targeted congenital CMV (cCMV) screening in newborns who failed universal newborn hearing screening has been proposed as a strategy to identify neonates with both hearing loss and cCMV infection who could benefit from antiviral treatment implemented within the first month of life. To evaluate the feasibility and performance of cCMV targeted screening in a French setting. Neonates were recruited in 5 maternity centers in greater Paris. A saliva sample for CMV polymerase chain reaction (PCR) testing was collected in neonates who failed newborn hearing screening. Outcomes including CMV PCR result and confirmation of hearing loss by an otorhinolaryngologist specialist were documented. Two-hundred thirty-six newborns were included and a saliva sample was collected in 98% (231/236) of them. The result of CMV PCR was available at a median of 9 days (7-10 days) of life and in 96% of cases within the first month of life. Two neonates were infected with CMV. The result of the otorhinolaryngologist assessment was available in 75% (178/236) of cases at a median of 16 days (9-26 days). Hearing loss was confirmed in 2.8% (5/178). The 2 infected neonates had hearing loss confirmed at 5 and 8 days of life and were treated with valganciclovir at days 9 and 16, respectively. The result of this study confirms that targeted cCMV screening is feasible in these French settings.
